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Education

Licensure

Electronics Engineer – PRC License No. 0070032 (May 2018)

Electronics Technician – PRC License No. 0016752 (May 2018)

Collegiate Level

Bachelor of Science in Electronics Engineering

Mapua University

Muralla St., Intramuros, Manila, Philippines

July 2012 - March 2018

Objective

A Full Stack Developer and IT Automation Engineer with extensive experience in designing, developing, and maintaining manufacturing web applications and industrial automation systems. Currently working at Taiyo Yuden Co. Ltd. in Japan, seeking a new opportunity to leverage software development and automation expertise in a dynamic IT environment.

Technical Skills

- **Full Stack Development:** HTML, CSS, JavaScript, PHP (MVC Architecture), React, .NET 9 / C#, MAUI Android
- **Database:** MySQL, SQL Server, Database Architecture Design
- **Industrial Automation:** PLC Programming (KV Studio – Keyence, CX Programmer – Omron, GX Developer – Mitsubishi, KGL – LG), HMI Programming (VT Studio, NB/NV Designer, CX Designer), SCADA (CX Supervisor), Servo/Stepper Motor Setup (Omron, Mitsubishi)
- **IoT & Hardware Integration:** RFID, Barcode Scanning (Zebra Devices), Sensor Systems (Proximity, Temperature, IR, Light)
- **Data Visualization:** Microsoft Power BI, QlikView
- **CAD:** AutoCAD 2D / Basic 3D, Electrical Diagram Design
- **Networking & Security:** Computer Network Systems, Advanced Networking Protocols, Network Security, Malware Prevention
- **Other Programming:** Python, C++, Macro VBA Scripting
- **Tools & Platforms:** Git, IIS, Windows Server, MySQL Workbench, Microsoft Office (Word, Excel, PowerPoint)

Work Experience

Facility Development Engineer at **Taiyo Yuden Co. Ltd.** (Tamamura, Gunma, Japan)

Nov 2023 – Present (**2 years & 7 months**)

Full Stack Developer

- Independently designed, developed, and maintained multiple manufacturing web applications from the ground up, handling the full software development lifecycle — from requirements gathering, system architecture, development, testing, deployment, and ongoing maintenance.
- TRS (Training Record System) — A multi-site web-based training management system built in PHP using MVC architecture, deployed across 4 manufacturing sites (JTY, JNTY, JCTY, MSTY), serving 6,000+ users. The system manages employee training schedules, progress tracking, training history records, and completion status. Supports multi-modal attendance logging — via manual login, barcode scanning, or RFID tap — enabling seamless and flexible attendance capture across all sites.
- IMS (Inventory Management System) — A full stack inventory tracking system consisting of a .NET 9 REST API backend, a React web application, and a MAUI Android app integrated with Zebra barcode scanning devices for real-time inventory tracking on the shop floor.
- Machine Monitoring Systems — Developed real-time web-based monitoring dashboards for production equipment, providing visibility into machine production data and output per machine and shift.
- Administered and designed database architecture across all managed systems using MySQL Workbench and SQL Server.

Automation

- Designed and built automated machines from scratch, handling the complete automation workflow including PLC programming using KV Studio (Keyence), electrical wiring diagram design using 2D CAD, HMI/touch panel programming using VT Studio, and actual electrical wiring and hardware setup.
- Utilized IoT technology to monitor and visualize the operation status and output accuracy of production equipment, extract process issues, and implement countermeasures.

***Senior Specialist, Master Data at Henkel Pacific Asia Service Centre (Makati City, Philippines)
April 2023 – October 2023 (7 months)***

- Responsible for generating International Packaging Specification (IPAS) data and executing master data clean up as required by the business
- Responsible for ensuring that IPAS data are free from inconsistencies / errors. Mitigating any findings on timely basis
- Responsible for various coordination in gathering the information needed for IPAS corrections/adjustments; Include setting up meetings, sending queries/clarifications to relevant parties, and communication with supplier for gathering related IPAS information

Test Equipment Engineer (Systems and Automation) at STMicroelectronics (Calamba, Laguna, Philippines) May 2021 – March 2023 (1 year & 10 months)

System Engineer: Equipment Maintenance and Control System (EMCS)

- Provided system support in EMCS to Operations 2 Test and Finish Area, covering both hardware and dashboard.
- Responsible for the equipment setup of MicroTerminal/Selesta installation for EMCS, including LAN network activation, IP address assignment, and machine pairing and interfacing.
- Managed machine groupings for corresponding products in the EMCS Dashboard and configured timing settings per machine grouping.
- Encoded defined error codes corresponding to each machine grouping.
- Maintained and edited EMCS EyeView — a visual representation of shop floor machines showing their location, status, output quantity, efficiency, and UPH.
- Supported the maintenance of Net Equipment Efficiency (NEE) at 90% across all machines in the Test and Finish Area. Participated in projects to improve efficiency, including OEU implementation to identify and address DC shortstops causing NEE drops.
- Led a project to upgrade all HDDs to SSDs across machine systems to improve boot time and maximize equipment output. Managed machine software backups to both physical drives and network storage.

Network Security

- Responsible for malware detection across all machine equipment in the Test and Finish Area.
- Investigated and resolved malware and virus incidents and implemented preventive measures to avoid recurrence.

- Provided McAfee RMR Sealing support — controlled applications that posed infection risks and raised tickets in Helix for RMR Sealing updates and mode changes.
- Supported software-related activities including upgrades and modifications.

Automation

- Developed a QlikView Net Equipment Efficiency Dashboard for Operations 2 Test and Finish Area, including data analysis, computation logic, and continuous improvement initiatives. Interfaced with the DIT Team for EMCS data and queries.
- Contributed to the development of an NEE Dashboard in Microsoft Power BI.
- Developed automation scripts to streamline operations, including auto-delete and auto-backup scripts triggered when HDD/SSD capacity reached 80%, transferring files to a shared network drive.

Automation Engineer at Nexgen Technology Inc. (Las Piñas City, Philippines) August 2018 – May 2021 (2yrs & 8 months)

- Designed PLC/HMI programs for machine handlers used in semiconductor manufacturing companies (Continental Philippines, Vishay Philippines, ONSem, etc.) using CX Programmer for Omron PLCs, GX Developer for Mitsubishi PLCs, and KGL for LG PLCs.
- Deployed to semiconductor manufacturing plants to install and set up machine handlers and was responsible for monitoring and interfacing handlers with testers until ready for production.
- Served as a *Service Engineer* — on-call for troubleshooting machine handlers in a down state, including debugging PLC/HMI programs, analyzing electrical diagrams and manuals, and checking I/O signals to identify root causes and provide immediate solutions.
- Communicated directly with clients to discuss machine handler requirements, determine I/O count, and recommend cost-effective materials, PLCs, and sensors.
- Designed electrical diagrams using AutoCAD and produced electrical manuals documenting I/O connections. Responsible for electrical wiring and setup.
- Provided service and support for Baumann Handler installation, setup, and troubleshooting.
- Attended Baumann Automation General Training at Baumann GmbH, Amberg, Bayern, Germany (February 16 – March 17, 2019), covering Stäubli Robotic Arm Point Teaching (4-axis/6-axis), basic maintenance, and basic troubleshooting

Internship

Project based I.T. Support Intern at **Dimension Data Philippines Inc.** August 2017 – September 2017 (240hrs)

- Responsible for technical troubleshooting, installations and setup.
- Deployed at J&J and AXA to provide support in network installations.

Affiliations

Institute of Electronics and Communications Engineers of the Philippines (IECEP)

- ***Member:*** May 2018 – Present

Junior Philippine Computer Society – Mapua University (JPCS-MU)

- ***President:*** Jan 2016 – July 2016
- ***COO Representative:*** July 2015 – Jan 2016
- ***Project Committee Head:*** July 2015 – Jan 2016
- ***Internal Vice President:*** January 2015 – July 2015
- ***Membership Committee Head:*** July 2014 – Dec 2014
- ***Member:*** Jan 2014 – Present

Center for Student Advising – (CSA Mapua)

- ***Peer Adviser:*** May 2015 – December 2017